

ENTERPRISE RISK MANAGEMENT ANALYTICS

TD Bank Group Failed First Horizon Acquisition

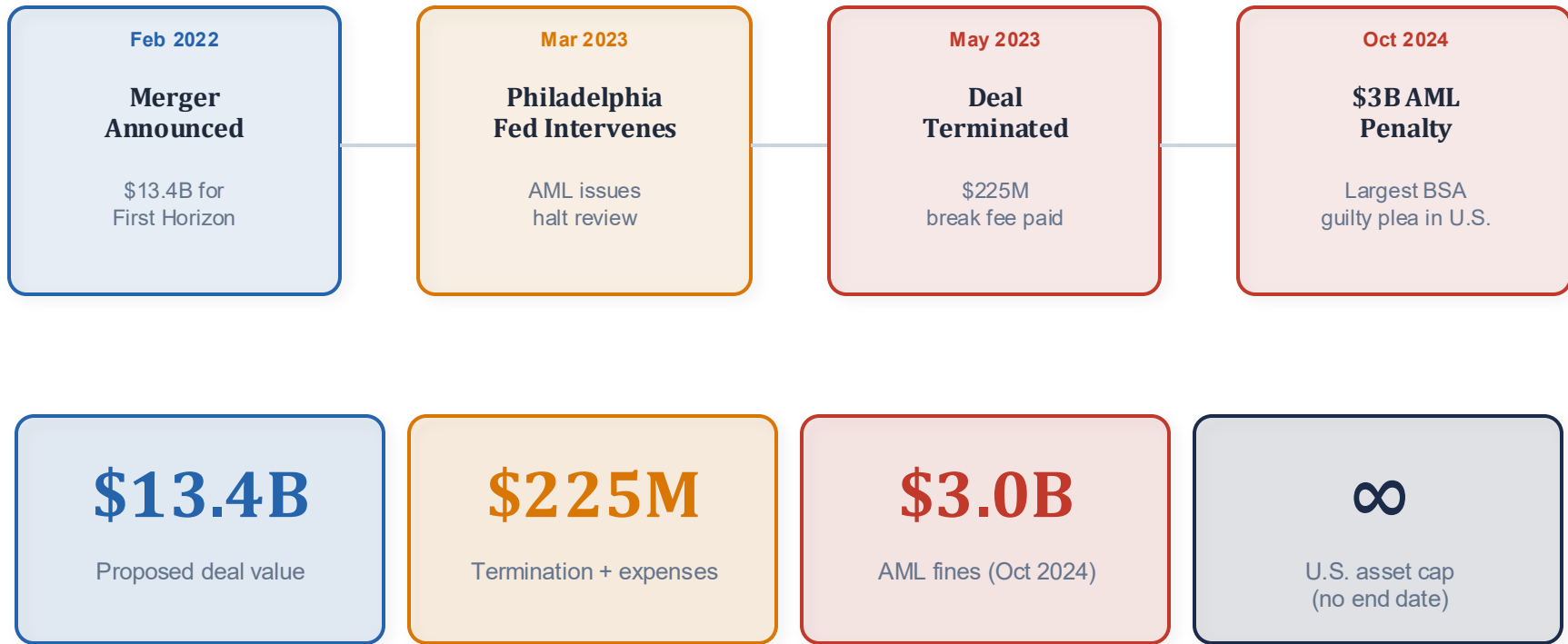
A Quantitative Enterprise Risk Assessment

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\$3.2B

The Deal That Never Closed



Three Enterprise Risks Identified

81/81

**Regulatory Approval
Failure**

▲ **CRITICAL**

TD's AML failures made regulatory approval impossible. \$3.225B in direct costs from termination fee and fines. Root cause of all subsequent risks.

63/81

**Market Share
Loss**

▲ **HIGH**

Asset cap freezes U.S. market share while competitors grow. TD lost planned 412 branches. Deposit growth: -6.0% YoY in 2024 vs. +3.1% industry.

63/81

**Reputational
Damage**

▲ **HIGH**

First U.S. bank to plead guilty to BSA violations. CEO resigned April 2025. Sustained crisis-level media coverage. Equity value at risk: ~\$7.7B.

Risks are positively correlated — AML failure triggered all three in sequence

Scenario Analytics & Industry Fusion

Scenario Analytics

Risk 1 — Regulatory:

Optimistic: AML remediation complete by 2025, cap lifted

Base: Cap persists through 2026, monitor oversight continues

Adverse: Additional violations found, enhanced penalties

Risk 2 — Market Share:

Base: TD loses 1–1.5pp market share over 3 years

Adverse: Net deposit outflows under prolonged cap

Risk 3 — Reputation:

Base: Recovery gradual, AMBER coverage through 2025

Adverse: Further executive departures sustain crisis

Industry Fusion Analytics

Key competitive findings:

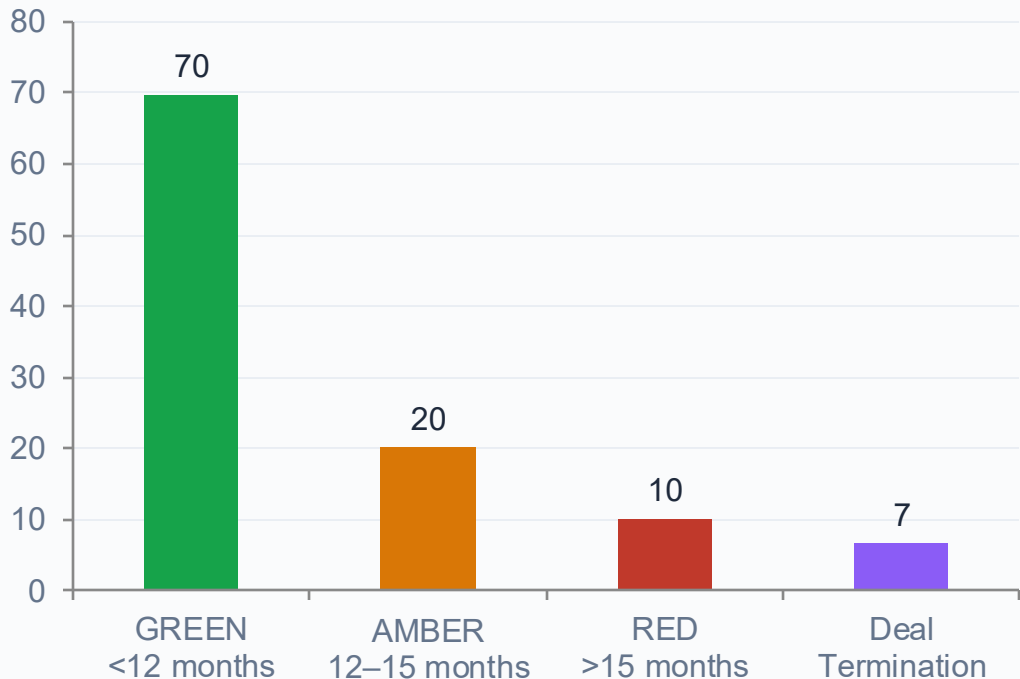
Post-2020 BSA/AML standards tightened. Competitors (JPMorgan, WF, BofA) had already modernized compliance infrastructure. TD had not.

FDIC industry avg deposit growth: 4.2% (2019–2023). TD 2024: -6.0%. A 10+ point divergence not explained by macro factors alone.

Fifth Third, Truist, and Regions Financial are positioned to capture TD's forfeited southeastern market share.

All 3 risks warranted quantitative analysis.

Risk 1: Regulatory Approval — Log-Normal Simulation



Distribution: Log-normal | $\mu = 10.8 \text{ mo}$, $\sigma = 3.2 \text{ mo}$ | $n = 10,000 \text{ iterations}$

10.8 mo

Industry mean
review duration

14.2 mo

TD Bank actual
(AMBER→RED)

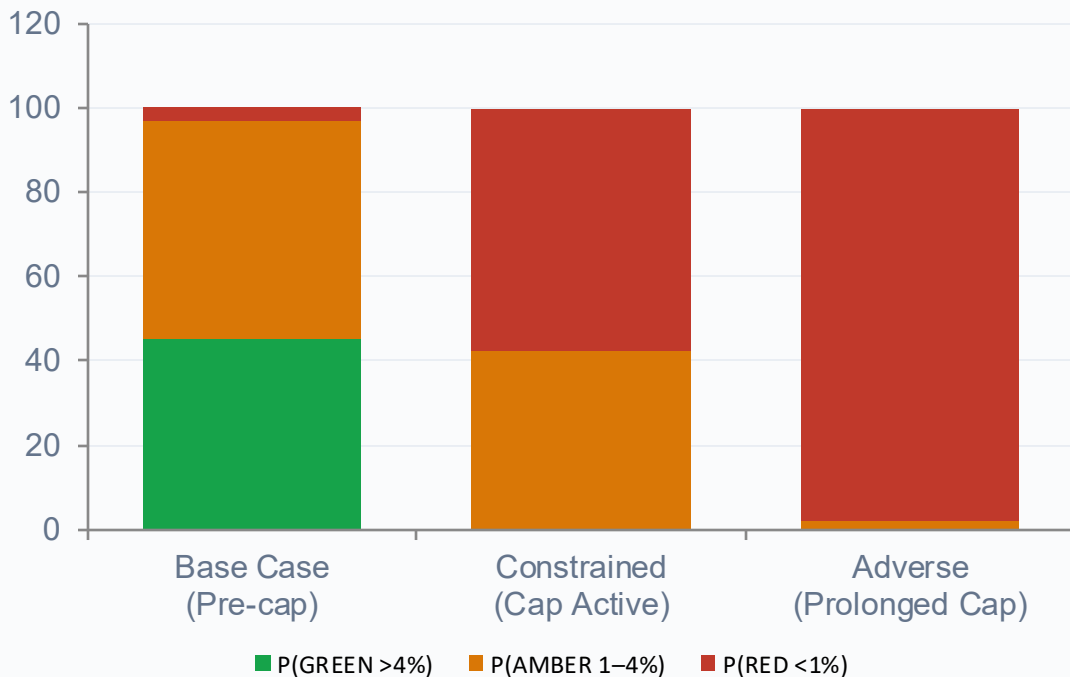
79th %ile

TD's position in
distribution

6.6%

Overall P(deal
termination)

Risk 2: Market Share Loss — Deposit Growth Scenarios



2024 Actual: -6.0% YoY

Deep RED — adverse scenario

FDIC industry avg: +3.1%

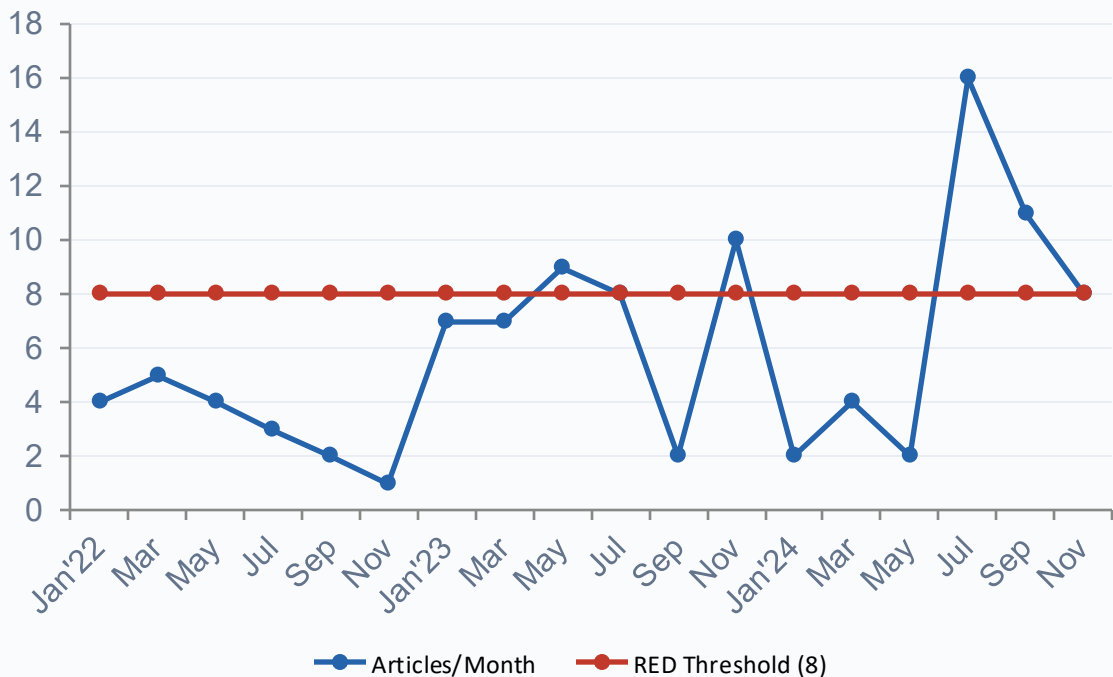
TD divergence: -9.1pp

Asset cap: No new branches,
no new product lines
without OCC approval

Expected Loss:
\$255M over 3 years

Parameters: Base $N(3.8\%, 1.5\%)$, Constrained $N(0.8\%, 1.0\%)$, Adverse $N(-2.0\%, 1.5\%)$ | FDIC + TD 40-F sources

Risk 3: Reputational Damage — Poisson Media Volume



Baseline ($\lambda=2$)

$P(\text{RED}) = 0.0\%$

Elevated ($\lambda=6$)

$P(\text{RED}) = 14.7\%$

Crisis ($\lambda=14$)

$P(\text{RED}) = 94.0\%$

6 RED-zone months observed | Oct 2024: $\lambda=16$ (peak crisis) | Stock sensitivity: -2% to -4% per RED month

Machine Learning: KRI-Based Zone Classification

82.1%

Random Forest CV accuracy
(Risk 2: deposit zone)

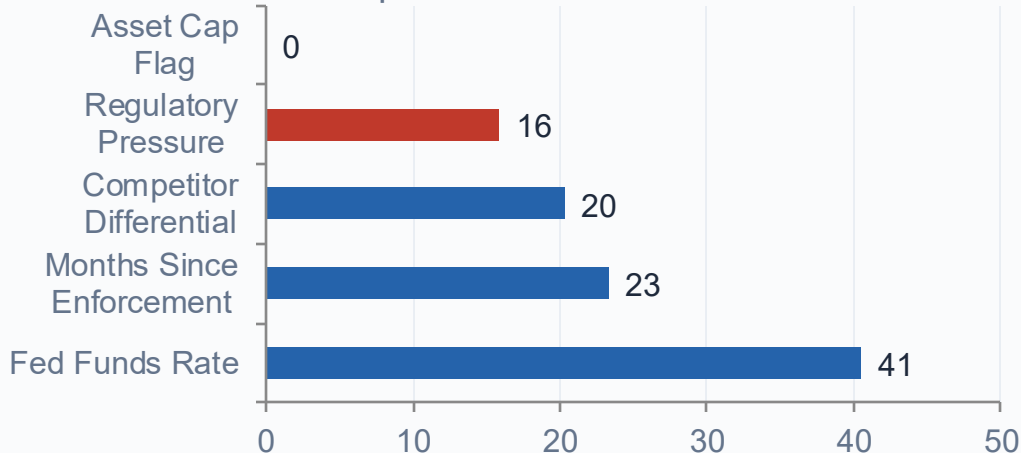
0.764

Gradient Boosting AUC
(Risk 3: RED zone)

$R^2=0.20$

Stock return ~ media volume
(Linear regression)

Feature Importances — Risk 2 Random Forest



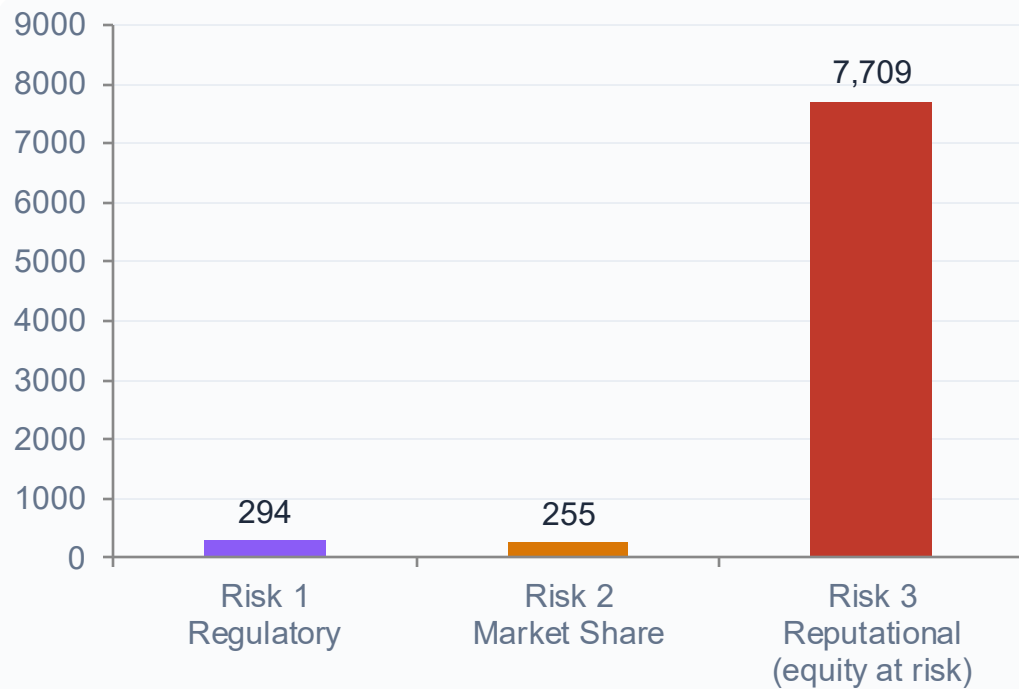
Key Insight

Fed Funds rate (macro) + months since enforcement (regulatory) together explain 63.8% of deposit zone prediction.

Deposit outcomes are jointly determined by interest rate conditions AND regulatory standing — neither alone is sufficient.

This validates using both business and non-business environment factors in the model.

Portfolio Expected Loss: ~\$8.26B USD



Expected Loss Calculations

Risk 1 $9.1\% \times \$3.225\text{B}$ **\$293.5M**

Risk 2 $85\% \times \$200\text{M} \times 1.5\text{pp}$ **\$255M**

Risk 3 $88\% \times 3\% \times \$73\text{B} \times 4\text{mo}$ **\$7.7B***

Total Portfolio estimate **~\$8.26B**

**Risk 3 = equity value at risk under crisis scenario, not a single direct loss*

KRI Dashboard — Current Status

Risk 1 — Regulatory Review Duration	GREEN < 12 months	AMBER 12–15 months	RED > 15 months	MONITORING N/A (post-termination)
Risk 2 — Market Share U.S. Deposit Growth YoY	GREEN > 4%	AMBER 1%–4%	RED < 1% or negative	RED -6.0% (2024 actual)
Risk 3 — Reputational Negative Media Articles/Month	GREEN < 3	AMBER 3–8	RED > 8	AMBER→RED ~8–10 (late 2024)

Risk Response Strategies

Risk 1

AVOID

No U.S. M&A until: (1) OCC cap lifted, (2) AML remediation independently certified, (3) 24-month clean regulatory window. Leverage DOJ monitor as credibility signal.

Risk 2

MITIGATE

Accelerate digital acquisition in existing Maine-Florida footprint. Quarterly deposit growth monitoring vs. FDIC. Engage OCC on cap removal timeline milestones. Deposit retention programs.

Risk 3

**MITIGATE +
TRANSFER**

Monthly media volume monitoring with 48-hour comms response protocol. Proactive quarterly investor updates on AML remediation progress. D&O insurance review. CEO transition positioned as compliance culture reset.

Conclusion

01

TD Bank's \$3.225B loss was not unforeseeable — the leading indicators were observable. The 14.2-month review sat at the 79th percentile; the ML classifier would have flagged the deposit zone shift in 2023.

02

Monte Carlo simulation shows P(RED) rises from 10% at baseline to 57-98% under current constrained conditions. The risk landscape has fundamentally shifted post-cap.

03

The KRI framework — review duration, deposit growth rate, media volume — provides the early warning infrastructure that was absent during the First Horizon process.

04

Total portfolio expected loss ~\$8.26B. The cost of reactive vs. proactive risk management is not academic — it is directly and quantifiably asymmetric.